

D866

General Chemistry II with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A cold pack sold by a sports-medicine supplier in boxes of 24 contains ammonium nitrate and a pouch of water. When the pouch breaks and the solid dissolves, the water temperature drops from 22.0 °C to 8.5 °C. How is the dissolving process classified?

- A. Endothermic; ΔH is positive.
- B. Endothermic; ΔH is negative.
- C. Neither; dissolving is a physical change, so no enthalpy change can occur.
- D. Exothermic; ΔH is negative, because the water lost heat.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A lab group heats 250.0 g of water in a 600 mL borosilicate beaker on a hot plate rated at 1,000 W. The temperature rises from 20.0 °C to 45.0 °C. Water's specific heat is 4.18 J/g·°C. How much heat did the water absorb?

- A. 47.0 kJ
- B. 1.05 kJ
- C. 1.50 kJ
- D. 26.1 kJ

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

2,090 J of heat is added to 50.0 g of water ($c = 4.18 \text{ J/g}\cdot^\circ\text{C}$). By how much does its temperature rise?

- A. It cannot be found without the starting temperature.
- B. 0.100 °C
- C. 10.0 °C
- D. 41.8 °C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A technician checking a chemical supplier's shipment finds a pressurized steel cylinder labeled only with the formula N_2O_4 , a rocket-propellant component that gives off a red-brown vapor. What is the correct systematic name to write on the inventory?

- A. Tetranitrogen dioxide
- B. Nitrogen oxide
- C. Nitrogen(IV) oxide
- D. Dinitrogen tetroxide

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An ammonia plant model uses the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, run at 450 °C and 200 atm with an iron catalyst. A reactor is charged with 2.0 mol N_2 and 4.5 mol H_2 . What is the maximum amount of NH_3 that can form?

- A. 4.0 mol
- B. 3.0 mol
- C. Less than 3.0 mol, because the iron catalyst is used up.
- D. 9.0 mol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, a mixture contains 6.0 mol H_2 and 4.0 mol O_2 . After the reaction goes to completion, how much of the excess reactant remains?

- A. 3.0 mol O_2
- B. None; every reactant is always used up completely.
- C. 2.0 mol H_2
- D. 1.0 mol O_2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A lab team calculates a theoretical yield of 12.0 g of lead(II) iodide. The product was a bright yellow precipitate, and after filtering through fluted paper and drying overnight they recover 9.6 g. What is the percent yield?

- A. 9.6%
- B. 80.0%
- C. 125%
- D. 20.0%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A synthesis gives 15.0 g of product at a 75.0% yield. What was the theoretical yield?

- A. 20.0 g
- B. 15.0 g, because theoretical and actual always match in a clean reaction.
- C. 60.0 g
- D. 11.3 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A prep technician dissolves 5.85 g of NaCl (58.44 g/mol) and dilutes it to 250.0 mL of solution. The balance was calibrated that morning, and the bottle is stored at room temperature on the reagent shelf. What is the molarity?

- A. 0.100 M
- B. 0.400 M
- C. 0.000400 M
- D. 2.50 M

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

How many grams of NaCl (58.44 g/mol) are needed to make 500.0 mL of 0.200 M solution?

- A. 5.84 g
- B. 0.100 g
- C. 11.7 g
- D. 5,844 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A technician mixes clear silver nitrate solution, bought from a photography supply company at \$38 per 100 g, with clear sodium chloride solution. A white solid forms at once. What is the solid?

- A.** Silver metal and chlorine gas
- B.** No solid can form, because both starting solutions are soluble.
- C.** Silver chloride, AgCl
- D.** Sodium nitrate, NaNO₃

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A lab team wants to test whether water temperature affects how fast sugar dissolves. They have a new set of digital thermometers donated by a local business and plenty of granulated sugar. Which plan gives a valid test?

- A.** Run one trial at each temperature, since repeated trials only add error.
- B.** Use the most precise donated thermometers and skip controlling the sugar mass.
- C.** Use equal masses of the same sugar and equal water volumes at several temperatures, stir the same way, and repeat each trial three times.
- D.** Stir the warmer beakers harder so the difference is easier to see.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A cold pack ~~sold by a sports medicine supplier in boxes of 24~~ contains ammonium nitrate and a pouch of water. When the pouch breaks and the solid dissolves, the water temperature drops from 22.0 °C to 8.5 °C. How is the dissolving process classified?

Target word: sign of ΔH for the dissolving system **Decides:** The surroundings (water) got colder, so the system absorbed heat.

A	Correct	Heat flowed from the water into the dissolving process, so the system gained energy.
B	Confusion set	Right label, wrong sign; endothermic means $\Delta H > 0$.
C	Absolute language	Physical changes such as dissolving and melting have enthalpy changes.
D	Inverted operation	Reads the sign from the surroundings instead of the system.

2. Answer D Concept 2

A lab group heats 250.0 g of water ~~in a 600 mL borosilicate beaker on a hot plate rated at 1,000 W~~. The temperature rises from 20.0 °C to 45.0 °C. Water's specific heat is 4.18 J/g·°C. How much heat did the water absorb?

Target word: heat absorbed **Decides:** $\Delta T = 45.0 - 20.0 = 25.0$ °C.

A	Wrong stem number	Uses the final temperature 45.0 °C instead of the change.
B	Dropped element	$250.0 \times 4.18 = 1,045$ J leaves out ΔT .
C	Inverted operation	$250.0 \times 25.0 / 4.18 \approx 1,495$ J divides by specific heat instead of multiplying.
D	Correct	$250.0 \times 4.18 \times 25.0 = 26,125$ J ≈ 26.1 kJ.

3. Answer C Concept 2 • transfer

2,090 J of heat is added to 50.0 g of water ($c = 4.18$ J/g·°C). By how much does its temperature rise?

Target word: solve for ΔT **Decides:** $\Delta T = q / (m \times c)$.

A	Adjacent category	Confuses the change in temperature with the final temperature; ΔT needs no starting value.
B	Inverted operation	$209 / 2,090$ flips the ratio.
C	Correct	$2,090 / (50.0 \times 4.18) = 2,090 / 209 = 10.0$ °C.
D	Dropped element	$2,090 / 50.0$ leaves out the specific heat.

4. Answer D Concept 3

A technician checking a chemical supplier's shipment finds a ~~pressurized steel cylinder~~ labeled only with the formula N_2O_4 , ~~a rocket propellant component that gives off a red brown vapor~~. What is the correct systematic name to write on the inventory?

Target word: name of a nonmetal-nonmetal compound **Decides:** Nitrogen and oxygen are both nonmetals, so prefixes are used.

A	Inverted operation	Swaps the subscripts between the two elements.
B	Dropped element	Leaves out the prefixes that tell the atom counts.
C	Adjacent category	Roman numerals belong to ionic compounds of metals with variable charge.
D	Correct	Two N (di-) and four O (tetra-); molecular compounds use prefixes.

5. Answer B Concept 4

An ammonia plant model uses the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, ~~run at 450 °C and 200 atm with an iron catalyst~~. A reactor is charged with 2.0 mol N_2 and 4.5 mol H_2 . What is the maximum amount of NH_3 that can form?

Target word: maximum product from the limiting reactant **Decides:** 2.0 mol N_2 needs 6.0 mol H_2 ; only 4.5 mol is present, so H_2 limits.

A	Confusion set	Treats N_2 as limiting because it has the smaller mole number.
B	Correct	$4.5 \text{ mol H}_2 \times (2 \text{ NH}_3 / 3 \text{ H}_2) = 3.0 \text{ mol NH}_3$.
C	Camouflage	A catalyst is not consumed and does not change the theoretical amount.
D	Dropped element	Doubles 4.5 mol H_2 and ignores the 3:2 ratio.

6. Answer D Concept 4 • transfer

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, a mixture contains 6.0 mol H_2 and 4.0 mol O_2 . After the reaction goes to completion, how much of the excess reactant remains?

Target word: leftover excess reactant **Decides:** 6.0 mol H_2 uses 3.0 mol O_2 .

A	Dropped element	That is the O_2 used, not the O_2 left over.
B	Absolute language	Reactants in a non-ratio mixture leave an excess.
C	Inverted operation	Subtracts 4.0 from 6.0 as if they reacted 1:1, and names the wrong excess.
D	Correct	$4.0 - 3.0 = 1.0 \text{ mol O}_2$ left; H_2 is limiting.

7. Answer B Concept 5

A lab team calculates a theoretical yield of 12.0 g of lead(II) iodide. ~~The product was a bright yellow precipitate, and after filtering through fluted paper and drying overnight they recover 9.6 g.~~ What is the percent yield?

Target word: percent yield **Decides:** Actual 9.6 g over theoretical 12.0 g.

A	Wrong stem number	Reports the recovered mass as a percent.
B	Correct	$9.6 / 12.0 \times 100 = 80.0\%$.
C	Inverted operation	$12.0 / 9.6$ divides theoretical by actual.
D	Adjacent category	That is the percent lost, not the yield.

8. Answer A Concept 5 • transfer

A synthesis gives 15.0 g of product at a 75.0% yield. What was the theoretical yield?

Target word: solve for theoretical yield **Decides:** Theoretical = actual / (percent yield / 100).

A	Correct	$15.0 / 0.750 = 20.0$ g.
B	Absolute language	A yield under 100% means the theoretical amount is larger.
C	Confusion set	$15.0 / 0.250$ uses the percent lost.
D	Inverted operation	15.0×0.750 multiplies instead of dividing.

9. Answer B Concept 6

A prep technician dissolves 5.85 g of NaCl (58.44 g/mol) and dilutes it to 250.0 mL of solution. ~~The balance was calibrated that morning, and the bottle is stored at room temperature on the reagent shelf.~~ What is the molarity?

Target word: molarity **Decides:** $5.85 \text{ g} / 58.44 \text{ g/mol} = 0.100 \text{ mol}$; $250.0 \text{ mL} = 0.2500 \text{ L}$.

A	Dropped element	Reports moles and never divides by volume.
B	Correct	$0.100 \text{ mol} / 0.2500 \text{ L} = 0.400 \text{ M}$.
C	Wrong stem number	Divides by 250.0 as if milliliters were liters.
D	Inverted operation	$0.2500 \text{ L} / 0.100 \text{ mol}$ flips the ratio.

10. Answer A Concept 6 • transfer

How many grams of NaCl (58.44 g/mol) are needed to make 500.0 mL of 0.200 M solution?

Target word: solve for solute mass **Decides:** Moles = $0.200 \text{ M} \times 0.5000 \text{ L} = 0.100 \text{ mol}$.

A	Correct	$0.100 \text{ mol} \times 58.44 \text{ g/mol} = 5.84 \text{ g}$.
B	Confusion set	Stops at moles and labels them as grams.
C	Dropped element	0.200×58.44 ignores the half-liter volume.
D	Wrong stem number	Uses 500.0 mL as 500.0 L.

11. Answer C Concept 7

A technician mixes clear silver nitrate solution, ~~bought from a photography supply company at \$38 per 100 g,~~ with clear sodium chloride solution. A white solid forms at once. What is the solid?

Target word: identity of the precipitate **Decides:** Chlorides of silver are insoluble; nitrates and sodium salts are soluble.

A	Adjacent category	Treats a double-replacement reaction as a redox reaction.
B	Absolute language	Soluble reactants can swap partners to form an insoluble product.
C	Correct	$\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)}$; Na^+ and NO_3^- are spectators.
D	Confusion set	Nitrates and sodium salts are soluble; this stays dissolved.

12. Answer C Concept 8

A lab team wants to test whether water temperature affects how fast sugar dissolves. They have ~~a new set of digital thermometers donated by a local business~~ and plenty of granulated sugar. Which plan gives a valid test?

Target word: valid experimental design **Decides:** Only temperature should change; everything else stays the same, with repeated trials.

A	Inverted operation	Repetition reduces the effect of random error; it does not add it.
B	Camouflage	Better thermometers do not fix an uncontrolled variable.
C	Correct	One independent variable, controlled conditions, and replication.
D	Adjacent category	Adds a second variable (stirring), so the cause cannot be separated.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Endothermic vs. exothermic sign	ΔH is positive when the system absorbs heat (surroundings cool) and negative when it releases heat (surroundings warm).	Reading the sign from the water or the room instead of from the reacting system.	A temperature change in the water or container around a process.
2	Calorimetry: $q = mc\Delta T$	Heat equals mass $\times$ specific heat $\times$ temperature change, where $\Delta T = \text{final} - \text{initial}$.	Using the final temperature instead of the change, or dropping one factor.	A mass, a specific heat, and two temperatures.
3	Naming molecular compounds	Two nonmetals use Greek prefixes (mono-, di-, tri-, tetra-); Roman numerals are only for metals with more than one ion charge.	Using a Roman numeral or dropping the prefixes on a nonmetal-nonmetal compound.	A formula made only of nonmetals.
4	Limiting reactant	The limiting reactant is the one that runs out first based on the mole ratio, not the one with the smaller number of moles.	Picking the reactant with the smaller mole amount or mass.	Amounts are given for two reactants.
5	Percent yield	Percent yield = actual yield / theoretical yield $\times$ 100.	Dividing theoretical by actual, or reporting the percent lost.	"Theoretical" and "actual" (or "recovered") masses in one stem.
6	Molarity from mass	Molarity = moles of solute / liters of solution; convert grams to moles and mL to L first.	Dividing grams by volume or using milliliters as liters.	A solute mass and a solution volume in mL.
7	Predicting precipitates	In a double-replacement reaction, the product that is insoluble by the solubility rules forms the solid; nitrates and alkali-metal salts stay dissolved.	Thinking two soluble reactants cannot form a solid, or picking the spectator salt.	Two aqueous ionic solutions are mixed.
8	Controlled experimental design	Change one independent variable, keep the rest constant, and run repeated trials.	Changing two things at once to make the effect bigger.	"Design a test of whether X affects Y."

Course scope (public description): Chemistry II with Lab course is designed to equip students with a comprehensive understanding of advanced chemical principles and their real-world applications. Covering a wide range of topics from energy transfer during physical and chemical processes to the recognition and naming of compounds, the course emphasizes critical thinking and practical skills. Students will delve into quantitative relationships in chemical reactions, gaining proficiency in stoichiometry, solution concentration, and reaction predictions. Through scientific inquiry, students will learn to design experiments, collect and analyze data, and apply their knowledge to solve environmental and industrial problems. The course also includes extensive practice in communicating scientific findings effectively, both in written reports and oral presentations. With a blend of theoretical knowledge and hands-on laboratory experience, this course prepares students for success in advanced studies and professional applications of chemistry.